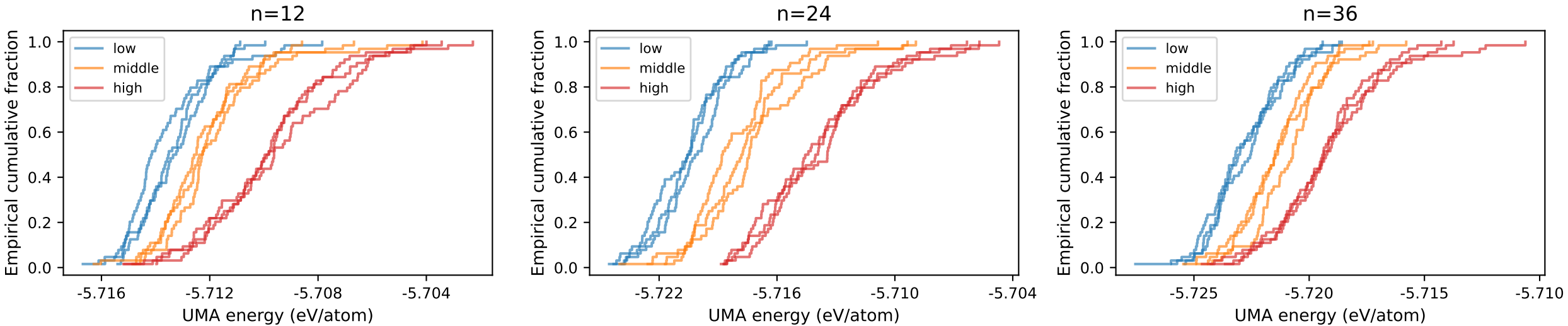mutation; thin curves are design replicates